

SOLIDWORKS API (.NET VBA C#) — Reference (299/337): ISurface (part 2 of 2)

SOLIDWORKS API Documentation (converted from CHM)

ISurface (part 2 of 2) (Type)

SOLIDWORKS API Help

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IGetBSurfParams Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetBSurfParams Method (ISurface)

Gets the b-spline surface parameters for the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBSurfParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double
```

```
value = instance.IGetBSurfParams()
```

C#

```
System.double IGetBSurfParams()
```

C++/CLI

```
System.double IGetBSurfParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

For COM applications, the following values must be set up in a previous call to [ISurface::IGetBSurfParamsSize3](#):

- vP0 [0] & vP0 [2] are the lower bounds of the U & V surface parameters, respectively.
- vP0 [1] & vP0 [3] are the upper bounds of the U & V surface parameters, respectively.

If you want to use the Bsurface in combination with its trim curves, you should use the [IFace2::GetTrimCurves2](#) method to extract the trim curve: and the Bsurface. The IFace2::GetTrimCurves method provides better alignment of the trim curves with the Bsurface because they are both generated at the same time.

Evaluation is as follows:

Number of returned double elements is (4 + NumUKnots + NumVKnots + NumControlPointDoubles), where NumUKnots = (NumColumnsControlPoints + Uorder) and NumVKnots = (NumRowsControlPoints + Vorder).

The returned data are arranged in the following order:

- Uorder, Vorder - These are two ints packed into a double
 - Uorder - Order of surface in U direction.
 - Vorder - Order of surface in V direction.
- NumColumnsControlPoints, NumRowsControlPoints - These are two ints packed into a double
 - NumColumnsControlPoints - Number of columns for the control points.
 - NumRowsControlPoints - Number of rows for the control points.
- Uperiodicity, Vperiodicity - These are two ints packed into a double. Because you may be generating a Bsurface with this function, the U,V periodicity values may be opposite from the original surface. See [ISurface::Parameterization](#) or [ISurface::IParameterization](#).
 - Uperiodicity - TRUE if surface is periodic in U direction.
 - Vperiodicity - TRUE if surface is periodic in V direction.
- DimensionControlPoints, dummy - These are two ints packed into a double
 - DimensionControlPoints - Dimension for control points.

- DimensionControlPoints = 3 for non-rational surfaces and control point coordinates are output as [x0,y0,z0,x1,y1,z1,.....].
- DimensionControlPoints = 4 for rational surfaces and control point coordinates and the weight are output as [x0,y0,z0,w0,x1,y1,z1,w1,.....].
- dummy - a system used filler.
- UKnots - Knot vector in the u direction
There will be (NumColumnsControlPoints + Uorder) knot values. If the surface is periodic in the U direction, then data will be converted and returned in a non-periodic form with additional knots added to the surface ends.
- VKnots - Knot vector in the v direction.
There will be (NumRowsControlPoints + Vorder) knot values. If the surface is periodic in the V direction, then data will be converted and returned in a non-periodic form with additional knots added to the surface ends.
- ControlPoints - This is a list of control points output in (NumColumnsControlPoints*NumRowsControlPoints) vectors of dimension DimensionControlPoints. The vectors are output row by row. The U direction of surface is in the row direction. The V direction of surface is in the column direction.
- For non-rational surfaces, control point coordinates are output as [x0,y0,z0,x1,y1,z1,.....].
- For rational surfaces, control point coordinates and the weight are output as [x0,y0,z0,w0,x1,y1,z1,w1,.....].

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::GetBSurfParams2 Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetBSurfParamsSize Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetBSurfParamsSize Method (ISurface)

Obsolete. Superseded by [ISurface::IGetBSurfParamsSize3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBSurfParamsSize( _
    ByVal WantCubicRational As System.Boolean, _
    ByRef Range As System.Double _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISurface
Dim WantCubicRational As System.Boolean
Dim Range As System.Double
Dim value As System.Integer

value = instance.IGetBSurfParamsSize(WantCubicRational, Range)
```

C#

```
System.int IGetBSurfParamsSize(
    System.bool WantCubicRational,
    ref System.double Range
)
```

C++/CLI

```
System.int IGetBSurfParamsSize(
    & System.bool WantCubicRational,
    & System.double% Range
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubicRational

Range

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetBSurfParamsSize](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetBSurfParamsSize2 Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetBSurfParamsSize2 Method (ISurface)

Obsolete. Superseded by [ISurface::IGetBSurfParamsSize3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBSurfParamsSize2( _  
    ByVal WantCubic As System.Boolean, _  
    ByVal WantNonRational As System.Boolean, _  
    ByRef Range As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim WantCubic As System.Boolean  
Dim WantNonRational As System.Boolean  
Dim Range As System.Double  
Dim value As System.Integer  
  
value = instance.IGetBSurfParamsSize2(WantCubic, WantNonRational, Range)
```

C#

```
System.int IGetBSurfParamsSize2(  
    System.bool WantCubic,  
    System.bool WantNonRational,  
    ref System.double Range  
)
```

C++/CLI

```
System.int IGetBSurfParamsSize2(  
& System.bool WantCubic,  
& System.bool WantNonRational,  
& System.double% Range  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

WantNonRational

Range

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetBSurfParamsSize2](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetBSurfParamsSize3 Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetBSurfParamsSize3 Method (ISurface)

Gets the allocation size necessary for Bsurface parameter data retrieval in a subsequent call to [ISurface::IGetBSurfParams](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetBSurfParamsSize3( _  
    ByVal WantCubic As System.Boolean, _  
    ByVal WantNonRational As System.Boolean, _  
    ByRef Range As System.Double, _  
    ByVal Tolerance As System.Double, _  
    ByRef Sense As System.Boolean _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim WantCubic As System.Boolean  
Dim WantNonRational As System.Boolean  
Dim Range As System.Double  
Dim Tolerance As System.Double  
Dim Sense As System.Boolean  
Dim value As System.Integer  
  
value = instance.IGetBSurfParamsSize3(WantCubic, WantNonRational, Range, Tolerance, Sense)
```

C#

```
System.int IGetBSurfParamsSize3(  
    System.bool WantCubic,  
    System.bool WantNonRational,  
    ref System.double Range,  
    System.double Tolerance,  
    out System.bool Sense  
)
```

C++/CLI

```
System.int IGetBSurfParamsSize3(  
& System.bool WantCubic,  
& System.bool WantNonRational,  
& System.double% Range,  
& System.double Tolerance,  
& [Out] System.bool Sense  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WantCubic

True for surface to be a cubic, false for not

WantNonRational

True if non-rational is needed, false if not; specifying true converts any surface type to a non-rational Bsurface; if you specify true, then you should only use this method for surfaces that are Bsurface or blend surface; otherwise, the underlying call is not made and the values returned from this are not initialized or contain the values from the last call

Range

Array of 4 doubles describing the U,V Range

Tolerance

Tolerance, in meters, between the approximated b-spline surface and the underlying surface; the default value is 0.01 and should generally be reduced to the tolerance desired

Sense

Approximated b-spline surface is not always in the same direction as the original surface; if sense is true, then the underlying surface and the b-spline surface are in the same direction

Return Value

Size of the data set

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetBSurfParamsSize3](#).

Remarks

Range contains the following values that can be obtained using [ISurface::Parameterization](#) or [ISurface::IParameterization](#):

- Range[0] & Range[2] are the lower bounds of the U & V surface parameters respectively.
- Range[1] & Range[3] are the upper bounds of the U & V surface parameters respectively.

See Also

[ISurface Interface](#)
[ISurface Members](#)

Availability

SOLIDWORKS 2000 SP3, Revision Number 8.3

SOLIDWORKS API Help

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IGetClosestPointOn Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetClosestPointOn Method (ISurface)

Uses the X,Y,Z input point to determine the closest point on the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetClosestPointOn( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
    ) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Double
```

```
value = instance.IGetClosestPointOn(X, Y, Z)
```

C#

```
System.double IGetClosestPointOn(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.double IGetClosestPointOn(  
& System.double X,
```

```
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

x coordinate of the approximate position on the surface

Y

y coordinate of the approximate position on the surface

Z

z coordinate of the approximate position on the surface

Return Value

- in-process, unmanaged C++: Pointer to an array of 5 doubles containing the (x,y,z) coordinates and the (u,v) parameters of the point on the surface
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::GetClosestPointOn Method](#)
[ISurface::GetProjectedPointOn Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetExtrusionsurfParams Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetExtrusionsurfParams Method (ISurface)

Gets the parameters for the extrusion surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetExtrusionsurfParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double
```

```
value = instance.IGetExtrusionsurfParams()
```

C#

```
System.double IGetExtrusionsurfParams()
```

C++/CLI

```
System.double IGetExtrusionsurfParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles giving extrusion surface parameters
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetExtrusionsurfParams](#).

Remarks

An extrusion surface is constructed by extruding a profile curve (line or arc or b-spline) along a direction vector. You can retrieve the profile curve for the extrusion using [ISurface::GetProfileCurve](#) or [ISurface::!GetProfileCurve](#).

Three doubles of parameters returned, *DirectionVector*[3], which is the direction vector along which the profile curve is extruded.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::GetExtrusionsurfParams Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMakeIsoCurvesCount Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetMakeIsoCurvesCount Method (ISurface)

Gets the number of curves that represent the ISO line of a given direction.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMakeIsoCurvesCount(  
    ByRef UvRange As System.Double, _  
    ByRef Dir As System.Double, _  
    ByVal Angle As System.Double, _  
    ByVal Tol As System.Double _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim UvRange As System.Double  
Dim Dir As System.Double  
Dim Angle As System.Double  
Dim Tol As System.Double  
Dim value As System.Integer  
  
value = instance.IGetMakeIsoCurvesCount(UvRange, Dir, Angle, Tol)
```

C#

```
System.int IGetMakeIsoCurvesCount(  
    ref System.double UvRange,  
    ref System.double Dir,  
    System.double Angle,  
    System.double Tol  
)
```

C++/CLI

```
System.int IGetMakeIsoCurvesCount(  
& System.double% UvRange,  
& System.double% Dir,  
& System.double Angle,  
& System.double Tol  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UvRange

Array of 4 doubles indicating the range of surface to use (see **Remarks**)

Dir

Array of 3 doubles indicating the direction of the projection on the surface (see **Remarks**)

Angle

Angle relative to Dir where to create the curves

Tol

Tolerance of the curves to create

Return Value

Number of curves to create

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetMakeIsoCurvesCount](#).

Remarks

The uvRange argument is an array of 4 doubles indicating the minimum and maximum U and V values:

[*u_min*, *u_max*, *v_min*, *v_max*]

The dir argument is an array of 3 doubles representing the unit vector:

[*x*, *y*, *z*]

Call this method before calling [ISurface::IMakeIsoCurves](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IMakeIsoCurve Method](#)
[ISurface::MakeIsoCurve Method](#)
[ISurface::MakeIsoCurves Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetOffsetSurfParams2 Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetOffsetSurfParams2 Method (ISurface)

Gets the overall offset distance of this offset surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetOffsetSurfParams2( _  
    ByRef BaseSurf As Surface, _  
    ByRef Sense As System.Boolean _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim BaseSurf As Surface  
Dim Sense As System.Boolean  
Dim value As System.Double
```

```
value = instance.IGetOffsetSurfParams2(BaseSurf, Sense)
```

C#

```
System.double IGetOffsetSurfParams2(  
    out Surface BaseSurf,  
    out System.bool Sense  
)
```

C++/CLI

```
System.double IGetOffsetSurfParams2(  
& [Out] Surface^ BaseSurf,  
& [Out] System.bool Sense  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BaseSurf

Base [surface](#) used to generate the offset surface

Sense

If true, then the offset is in the direction of the original surface's normal; if false, then the offset is in the opposite direction of the original surface's normal

Return Value

Offset for the surface

Visual Basic for Applications (VBA) Syntax

See [Surface.IGetOffsetSurfParams2](#).

Remarks

This method can get a pointer to the base surface and its sense. Both BaseSurf and Sense parameters can be a NULL pointer.

See Also

[ISurface Interface](#)

[ISurface Members](#)

[ISurface.IGetOffsetSurfParams2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetProfileCurve Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetProfileCurve Method (ISurface)

Gets the curve used to create the surface of revolution or extrusion surface and applies to only surface of revolution or extrusion surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetProfileCurve() As Curve
```

Visual Basic (Usage)



```
Dim instance As ISurface
```

```
Dim value As Curve
```

```
value = instance.IGetProfileCurve()
```

C#

```
Curve IGetProfileCurve()
```

C++/CLI

```
Curve^ IGetProfileCurve();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Profile [curve](#) used to create the surface of revolution or extrusion surface

Visual Basic for Applications (VBA) Syntax

See [Surface::IGetProfileCurve](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::GetProfileCurve Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetRevsurfParams Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IGetRevsurfParams Method (ISurface)

Gets the parameters for the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRevsurfParams() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double  
  
value = instance.IGetRevsurfParams()
```

C#

```
System.double IGetRevsurfParams()
```

C++/CLI

```
System.double IGetRevsurfParams();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

A surface of revolution is constructed by revolving a profile curve (line or arc or b-spline) about an axis. The profile curve for the surface of revolution can be retrieved with [ISurface::GetProfileCurve](#) or [ISurface::IGetProfileCurve](#).

Eight doubles of parameters returned.

- *Point [3]* - X, Y, Z coordinates of the start point of the axis of rotation.
- *AxisDir[3]* - Vector showing the direction of the axis of rotation.
- *Params[2]* - Parameter range of the part of the profile curve that was used to create the surface.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::GetRevsurfParams Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IntersectCurve Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectCurve Method (ISurface)

Obsolete. Superseded by [ISurface::IntersectCurve2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IIntersectCurve( _  
    ByVal OtherCurve As Curve, _  
    ByRef CurveBound As System.Double, _  
    ByVal PointCount As System.Integer, _  
    ByRef PointArray As System.Double, _  
    ByRef TArray As System.Double, _  
    ByRef UvArray As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherCurve As Curve  
Dim CurveBound As System.Double  
Dim PointCount As System.Integer  
Dim PointArray As System.Double  
Dim TArray As System.Double  
Dim UvArray As System.Double  
Dim value As System.Boolean  
  
value = instance.IIntersectCurve(OtherCurve, CurveBound, PointCount, PointArray, TArray, UvArray)
```

C#

```
System.bool IIntersectCurve(  
    Curve OtherCurve,  
    ref System.double CurveBound,  
    System.int PointCount,  
    out System.double PointArray,  
    out System.double TArray,  
    out System.double UvArray  
)
```

C++/CLI

```
System.bool IIntersectCurve(  
& Curve^ OtherCurve,  
& System.double% CurveBound,  
& System.int PointCount,  
& [Out] System.double PointArray,  
& [Out] System.double TArray,  
& [Out] System.double UvArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherCurve

CurveBound

PointCount

PointArray

TArray

UvArray

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectCurve](#).

See Also

IntersectCurve2 Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectCurve2 Method (ISurface)

Gets a surface-curve intersection.

.NET Syntax

Visual Basic (Declaration)

```
Function IIntersectCurve2( _  
    ByVal OtherCurve As Curve, _  
    ByRef CurveBound As System.Double, _  
    ByVal PointCount As System.Integer, _  
    ByRef PointArray As System.Double, _  
    ByRef TArray As System.Double, _  
    ByRef UvArray As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherCurve As Curve  
Dim CurveBound As System.Double  
Dim PointCount As System.Integer  
Dim PointArray As System.Double  
Dim TArray As System.Double  
Dim UvArray As System.Double  
Dim value As System.Boolean
```

```
value = instance.IIntersectCurve2(OtherCurve, CurveBound, PointCount, PointArray, TArray, UvArray)
```

C#

```
System.bool IIntersectCurve2(  
    Curve OtherCurve,  
    ref System.double CurveBound,  
    System.int PointCount,  
    out System.double PointArray,  
    out System.double TArray,  
    out System.double UvArray  
)
```

C++/CLI

```
System.bool IIntersectCurve2(  
& Curve^ OtherCurve,  
& System.double% CurveBound,  
& System.int PointCount,  
& [Out] System.double PointArray,  
& [Out] System.double TArray,  
& [Out] System.double UvArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherCurve

[Curve](#)

CurveBound

Array of 6 doubles representing the start and end points of the curve

PointCount

Number of points

PointArray

Array of points of size PointCount*3

TArray

Array of parameters on curve of size PointCount

UvArray

Array of parameters on surface of size PointCount*2

Return Value

True if getting the intersection succeeded, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectCurve2](#).

Remarks

The curves must be bounded.

Before calling this method, call [ISurface::GetIntersectCurveCount2](#) to get the number of points for this surface-curve intersection.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IntersectCurve2 Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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IntersectSurface Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectSurface Method (ISurface)

Gets a surface-surface intersection.

.NET Syntax

Visual Basic (Declaration)

```
Function IIntersectSurface( _  
    ByVal OtherSurf As Surface, _  
    ByVal CurveCount As System.Integer, _  
    ByRef CurveArray As Curve, _  
    ByRef BoundsArray As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherSurf As Surface  
Dim CurveCount As System.Integer  
Dim CurveArray As Curve  
Dim BoundsArray As System.Double  
Dim value As System.Boolean  
  
value = instance.IIntersectSurface(OtherSurf, CurveCount, CurveArray, BoundsArray)
```

C#

```
System.bool IIntersectSurface(  
    Surface OtherSurf,  
    System.int CurveCount,  
    out Curve CurveArray,  
    out System.double BoundsArray  
)
```

C++/CLI

```
System.bool IIntersectSurface(  
& Surface^ OtherSurf,  
& System.int CurveCount,
```

```
& [Out] Curve^ CurveArray,  
& [Out] System.double BoundsArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherSurf

Intersecting [surface](#)

CurveCount

Number of curves

CurveArray

Array of curves of size CurveCount

BoundsArray

Array of [curve](#) bounds of size CurveCount*2

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectSurface](#).

Remarks

Before calling this method, call [ISurface::GetIntersectSurfaceCount](#) to get the curve count for this surface-surface intersection.

See Also

[ISurface Interface](#)

[ISurface Members](#)

[ISurface::IntersectSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMakeIsoCurve Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IMakeIsoCurve Method (ISurface)

Creates a curve that represents the ISO line of a given surface.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IMakeIsoCurve( _  
    ByVal UorV As System.Boolean, _  
    ByVal UvValue As System.Double _  
) As Curve
```

Visual Basic (Usage)

```
Dim instance As ISurface  
Dim UorV As System.Boolean  
Dim UvValue As System.Double  
Dim value As Curve
```

```
value = instance.IMakeIsoCurve(UorV, UvValue)
```

C#

```
Curve IMakeIsoCurve(  
    System.bool UorV,  
    System.double UvValue  
)
```



```
Curve^ IMakeIsoCurve(
& System.bool UorV,
& System.double UvValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UorV
True to specify V, false to specify U
UvValue
U or V vertex or point that specifies the intersection of two curves

Return Value

[Curve](#)

Visual Basic for Applications (VBA) Syntax

See [Surface::IMakeIsoCurve](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::MakeIsoCurve Method](#)
[ISurface::MakeIsoCurves Method](#)
[ISurface::IMakeIsoCurves Method](#)
[ISurface::IGetMakeIsoCurvesCount Method](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

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IMakeIsoCurves Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IMakeIsoCurves Method (ISurface)

Creates curves on a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IMakeIsoCurves( _
    ByRef UvRange As System.Double, _
    ByRef Dir As System.Double, _
    ByVal Angle As System.Double, _
    ByVal Tol As System.Double, _
    ByVal CurveCount As System.Integer, _
    ByRef Curves As Curve, _
    ByRef CurveBounds As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface
Dim UvRange As System.Double
Dim Dir As System.Double
Dim Angle As System.Double
Dim Tol As System.Double
Dim CurveCount As System.Integer
Dim Curves As Curve
Dim CurveBounds As System.Double
Dim value As System.Boolean
```

```
value = instance.IMakeIsoCurves(UvRange, Dir, Angle, Tol, CurveCount, Curves, CurveBounds)
```

C#

```
System.bool IMakeIsoCurves(  
    ref System.double UvRange,  
    ref System.double Dir,  
    System.double Angle,  
    System.double Tol,  
    System.int CurveCount,  
    out Curve Curves,  
    out System.double CurveBounds  
)
```

C++/CLI

```
System.bool IMakeIsoCurves(  
    & System.double% UvRange,  
    & System.double% Dir,  
    & System.double Angle,  
    & System.double Tol,  
    & System.int CurveCount,  
    & [Out] Curve^ Curves,  
    & [Out] System.double CurveBounds  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UvRange

Array of 4 doubles indicating the range of the surface to use (see **Remarks**)

Dir

Array of 3 doubles indicating the direction of projection on the surface (see **Remarks**)

Angle

Angle relative to dir where to create the curves

Tol

Tolerance of the curves

CurveCount

Number of curves to create (see **Remarks**)

Curves

Array of [curves](#) of size CurveCount

CurveBounds

Array of doubles of size 2*Curves indicating the parametric bounds of the curve

Return Value

True if curves are created on the surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IMakeIsoCurves](#).

Remarks

The UvRange argument is an array of 4 doubles indicating the minimum and maximum U and V values:

[*u_min*, *u_max*, *v_min*, *v_max*]

The dir argument is an array of 3 doubles representing the unit vector:

[*x*, *y*, *z*]

Before calling this method, call [ISurface::IGetMakeIsoCurvesCount](#).

See Also

[ISurface Interface](#)

[ISurface Members](#)

[ISurface::MakeIsoCurves Method](#)

[ISurface::MakeIsoCurve Method](#)

[ISurface::IMakeIsoCurve Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

IntersectCurve Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectCurve Method (ISurface)

Obsolete. Superseded by [ISurface::IntersectCurve2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IntersectCurve( _  
    ByVal OtherCurve As System.Object, _  
    ByVal CurveBound As System.Object, _  
    ByRef PointArray As System.Object, _  
    ByRef TArray As System.Object, _  
    ByRef UvArray As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherCurve As System.Object  
Dim CurveBound As System.Object  
Dim PointArray As System.Object  
Dim TArray As System.Object  
Dim UvArray As System.Object  
Dim value As System.Boolean  
  
value = instance.IntersectCurve(OtherCurve, CurveBound, PointArray, TArray, UvArray)
```

C#

```
System.bool IntersectCurve(  
    System.object OtherCurve,  
    System.object CurveBound,  
    out System.object PointArray,  
    out System.object TArray,  
    out System.object UvArray  
)
```

C++/CLI

```
System.bool IntersectCurve(  
& System.Object^ OtherCurve,  
& System.Object^ CurveBound,  
& [Out] System.Object^ PointArray,  
& [Out] System.Object^ TArray,  
& [Out] System.Object^ UvArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherCurve

CurveBound

PointArray

TArray

UvArray

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectCurve](#).

See Also

IntersectCurve2 Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectCurve2 Method (ISurface)

Gets a surface-curve intersection.

.NET Syntax

Visual Basic (Declaration)

```
Function IntersectCurve2( _  
    ByVal OtherCurve As System.Object, _  
    ByVal CurveBound As System.Object, _  
    ByRef PointArray As System.Object, _  
    ByRef TArray As System.Object, _  
    ByRef UvArray As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherCurve As System.Object  
Dim CurveBound As System.Object  
Dim PointArray As System.Object  
Dim TArray As System.Object  
Dim UvArray As System.Object  
Dim value As System.Boolean  
  
value = instance.IntersectCurve2(OtherCurve, CurveBound, PointArray, TArray, UvArray)
```

C#

```
System.bool IntersectCurve2(  
    System.object OtherCurve,  
    System.object CurveBound,  
    out System.object PointArray,  
    out System.object TArray,  
    out System.object UvArray  
)
```

C++/CLI

```
System.bool IntersectCurve2(  
& System.Object^ OtherCurve,  
& System.Object^ CurveBound,  
& [Out] System.Object^ PointArray,  
& [Out] System.Object^ TArray,  
& [Out] System.Object^ UvArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherCurve

[Curve](#)

CurveBound

Array of 6 doubles representing the start and end points of the curve

PointArray

Array of points

TArray

Array of parameters on curve

UvArray

Array of parameters on surface

Return Value

True if getting the intersection succeeded, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectCurve2](#).

Remarks

The curves must be bounded.

Visual Basic and C++ Dispatch applications should use this method instead of using [ISurface::GetIntersectCurveCount2](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IntersectCurve2 Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IntersectSurface Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IntersectSurface Method (ISurface)

Gets a surface-surface intersection.

.NET Syntax

Visual Basic (Declaration)

```
Function IntersectSurface( _  
    ByVal OtherSurf As System.Object, _  
    ByRef CurveArray As System.Object, _  
    ByRef BoundsArray As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim OtherSurf As System.Object  
Dim CurveArray As System.Object  
Dim BoundsArray As System.Object  
Dim value As System.Boolean  
  
value = instance.IntersectSurface(OtherSurf, CurveArray, BoundsArray)
```

C#

```
System.bool IntersectSurface(  
    System.object OtherSurf,  
    out System.object CurveArray,  
    out System.object BoundsArray  
)
```

C++/CLI

```
System.bool IntersectSurface(  
& System.Object^ OtherSurf,  
& [Out] System.Object^ CurveArray,  
& [Out] System.Object^ BoundsArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

OtherSurf
Intersecting [surface](#)
CurveArray
Array of [curves](#)
BoundsArray

Array containing the curve bounds

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IntersectSurface](#).

Example

[Get Intersecting Faces \(VBA\)](#)

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IntersectSurface Method](#)
[ISurface::GetIntersectSurfaceCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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IParameterization Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IParameterization Method (ISurface)

Gets the parameterization of the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IParameterization() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double  
  
value = instance.IParameterization()
```

C#

```
System.double IParameterization()
```

C++/CLI

```
System.double IParameterization();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to an array of doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Surface::IParameterization](#).

Remarks

The return value is the following array of 11 doubles:

$[uRange[2], vRange[2], uBoundType, vBoundType, uProps[2], vProps[2], numProps]$

where:

uRange[2] = array of two doubles describing the low and high range, respectively, for U.

vRange[2] = array of two doubles describing the low and high range, respectively, for V.

uBoundType = two integers packed into a double. Each integer describes the behavior at the start and at the end of the U range. Valid values and their meanings are:

13733 - Infinite

13734 - Extendable

13735 - Not Extendable

13701 - Periodic

13736 - Periodic, but not continuously differentiable across the boundary

13741 - Degenerate

If the behavior at the start of the U range is Degenerate, then the V parameter is degenerate when U = *uRange*[0]. The derivative of the parameterization function with respect to v is 0 whenever u = *uRange*[0], for all values of v, and the parameter curve corresponding to u = *uRange*[0] degenerates to a single point. A parameterization can only degenerate at the end of its range, unless the surface is a type of Bsurface.

vBoundType = two integers packed into a double. Each integer describes the behavior at the start and end of the V range. For valid values and their meanings, see description of *uBoundType*.

uProps[2] = four integers packed into two doubles representing U parameterization properties. The *uProps* array of two doubles will contain a total of one or more integer values with a maximum of four. Valid values and their meanings are:

13701 - Periodic parameterization.

13737 - All derivatives continuous.

13738 - All derivatives non necessarily continuous.

13739 - Linear. Corresponding parameter is proportional to the distance along a straight line. Straight line corresponds to a constant value of the other parameter.

13740 - Circular. Corresponding parameter represents an angle around a circle. Circle corresponds to a constant value of the other parameter.

13746 - Bounds at the ends of the parameter range are coincident.

vProps[2] = four integers packed into two doubles representing V parameterization properties. The *vProps* array of two doubles will contain a total of one or more integer values with a maximum of four. For valid values and their meanings, see description of *uProps*.

numProps = two integers packed into a double. Each integer represents the number of properties returned in *uProps* and *vProps*, respectively.

If a surface is periodic in one direction (that is, cylinder, cone, torus (apple and lemon shapes), spheres), then U is the periodic direction.

The -10000 to 10000 parameter range is the modeling limit for the modeler. This means that the values are effectively infinite in size. To get the parameter range from a given face, use [IFace2::GetBox](#) or [IFace2::!GetBox](#), [ISurface::GetClosestPointOn](#) or [ISurface::!GetClosestPointOn](#), and [ISurface::ReverseEvaluate](#) or [ISurface::!ReverseEvaluate](#). These methods provide a true parameter range that the face is using on the surface.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::Parameterization Method](#)
[ISurface::!EvaluateAtPoint Method](#)
[ISurface::!Evaluate Method](#)
[ISurface::Evaluate Method](#)
[ISurface::EvaluateAtPoint Method](#)

SOLIDWORKS API Help

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IPlaneParams Property (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IPlaneParams Property (ISurface)

Gets the parameters of a planar surface.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IPlaneParams As System.Double

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double  
  
value = instance.IPlaneParams
```

C#

```
System.double IPlaneParams {get;}
```

C++/CLI

```
property System.double IPlaneParams {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the parameters of a planar surface

Visual Basic for Applications (VBA) Syntax

See [Surface::IPlaneParams](#).

Remarks

Returns an array of 6 double values:

normal.x
normal.y
normal.z
rootPoint.x
rootPoint.y
rootPoint.z

The *rootPoint* values are in meters.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::PlaneParams Property](#)
[ISurface::IsPlane Method](#)

SOLIDWORKS API Help

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IRverseEvaluate Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IRverseEvaluate Method (ISurface)

Gets the UV parameters at the specified location, which must be on the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IRverseEvaluate( _  
    ByVal PositionX As System.Double, _  
    ByVal PositionY As System.Double, _  
    ByVal PositionZ As System.Double _  
    ) As System.Double
```

Visual Basic (Usage)

```
Dim instance As ISurface  
Dim PositionX As System.Double  
Dim PositionY As System.Double
```



```
Dim PositionZ As System.Double
Dim value As System.Double

value = instance.IReverseEvaluate(PositionX, PositionY, PositionZ)
```

C#

```
System.double IReverseEvaluate(
    System.double PositionX,
    System.double PositionY,
    System.double PositionZ
)
```

C++/CLI

```
System.double IReverseEvaluate(
    & System.double PositionX,
    & System.double PositionY,
    & System.double PositionZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionX
X position on the surface

PositionY
Y position on the surface

PositionZ
Z position on the surface

Return Value

Array of doubles describing the UV parameters

Visual Basic for Applications (VBA) Syntax

See [Surface::IReverseEvaluate](#).

Remarks

At parametric singularities, such as sphere poles, the non-degenerate parameter is returned as the lowest value in its range. To determine the UV range, call [ISurface::Parameterization](#) or [ISurface::IParameterization](#).

[IFace2::ReverseEvaluate](#) and [IFace2::IReverseEvaluate](#) return corrected UV parameters for periodic surfaces; thus, you should use either of these methods when dealing with periodic surfaces.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::ReverseEvaluate Method](#)
[ISurface::Evaluate Method](#)
[ISurface::IEvaluate Method](#)
[ISurface::IEvaluateAtPoint Method](#)
[ISurface::EvaluateAtPoint Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsBlending Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsBlending Method (ISurface)

Gets whether the surface is a blending surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsBlending() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsBlending()
```

C#

```
System.bool IsBlending()
```

C++/CLI

```
System.bool IsBlending();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is a blending surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsBlending](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IBody2::CreateBlendSurface Method](#)
[IBody2::ICreateBlendSurface Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsCone Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsCone Method (ISurface)

Gets whether the surface is a cone.

.NET Syntax

Visual Basic (Declaration)

```
Function IsCone() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsCone()
```

C#

```
System.bool IsCone()
```

C++/CLI

```
System.bool IsCone();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is a cone, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsCone](#).

Example

[Locate Apex of Conical Face \(VBA\)](#)
[Get Parameters of Conical Surface \(VBA\)](#)
[Get Parameters of Conical Surface \(VB.NET\)](#)
[Get Parameters of Conical Surface \(C#\)](#)

Remarks

Closed cone faces that have an apex point do not have an edge or coedge at the apex.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface.IConeParams Property](#)
[ISurface.ConeParams Property](#)
[IModeler.CreateConicalSurface2 Method](#)
[IModeler.ICreateConicalSurface2 Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsCylinder Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsCylinder Method (ISurface)

Gets whether the surface is a cylinder.

.NET Syntax

Visual Basic (Declaration)

```
Function IsCylinder() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsCylinder()
```

C#

```
System.bool IsCylinder()
```

C++/CLI

```
System.bool IsCylinder();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is cylinder, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface.IsCylinder](#).

Example

[Calculate Closest Distance Between Faces \(VBA\)](#)
[Get Angle of Hole Not Normal to a Face \(VBA\)](#)
[Get Parameters of Cylindrical Surface \(VBA\)](#)

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IModeler.CreateCylindricalSurface2 Method](#)
[IModeler.ICreateCylindricalSurface2 Method](#)

IsForeign Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsForeign Method (ISurface)

Gets whether the surface is a foreign surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsForeign() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsForeign()
```

C#

```
System.bool IsForeign()
```

C++/CLI

```
System.bool IsForeign();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the surface is a foreign surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsForeign](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)

IsOffset Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsOffset Method (ISurface)

Gets whether the surface is an offset surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsOffset() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsOffset()
```

C#

```
System.bool IsOffset()
```

C++/CLI

```
System.bool IsOffset();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the surface is an offset surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsOffset](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[ISurface::IGetOffsetSurfParams2 Method](#)
[ISurface::GetOffsetSurfParams2 Method](#)
[ISurfaceOffsetFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsParametric Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsParametric Method (ISurface)

Gets whether the surface is a parametric (spline type) surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsParametric() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsParametric()
```

C#

```
System.bool IsParametric()
```

C++/CLI

```
System.bool IsParametric();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is a parametric surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsParametric](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)

SOLIDWORKS API Help

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ISphereParams Property (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : ISphereParams Property (ISurface)

Gets the parameters of a spherical surface.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ISphereParams As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double  
  
value = instance.ISphereParams
```

C#

```
System.double ISphereParams {get;}
```

C++/CLI

```
property System.double ISphereParams {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the parameters of a spherical surface

Visual Basic for Applications (VBA) Syntax

See [Surface::ISphereParams](#).

Remarks

Returns an array of 4 doubles:

center.x

center.y

center.z

radius

The *center* and *radius* values are in meters.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::SphereParams Property](#)
[ISurface::ISphere Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsPlane Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsPlane Method (ISurface)

Gets whether the surface is a planar surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsPlane() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface
Dim value As System.Boolean

value = instance.IsPlane()
```

C#

```
System.bool IsPlane()
```

C++/CLI

```
System.bool IsPlane();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is planar surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsPlane](#).

Example

[Create Infinite Plane \(VBA\)](#)
[Get Angle of Hole Not Normal to a Face \(VBA\)](#)
[Get Parameters of Planar Surface \(VBA\)](#)

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface.PlaneParams Property](#)
[ISurface.IPlaneParams Property](#)
[IModeler.CreatePlanarSurface2 Method](#)
[IModeler.ICreatePlanarSurface2 Method](#)
[ISurfacePlanarFeatureData Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsRevolved Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsRevolved Method (ISurface)

Gets whether the surface is a revolved surface.

.NET Syntax

Visual Basic (Declaration)

```
Function IsRevolved() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface
Dim value As System.Boolean

value = instance.IsRevolved()
```

C#

```
System.bool IsRevolved()
```

C++/CLI

```
System.bool IsRevolved();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the surface is a revolved surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsRevolved](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurfRevolveFeatureData Interface](#)
[IFeatureManager::InsertRevolvedRefSurface Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsSphere Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsSphere Method (ISurface)

Gets whether the surface is a sphere.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSphere() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsSphere()
```

C#

```
System.bool IsSphere()
```

C++/CLI

```
System.bool IsSphere();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is spherical, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsSphere](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[ISurface::ISphereParams Property](#)
[ISurface::SphereParams Property](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsSwept Method (ISurface)

[See Also](#)



Gets whether the surface is swept.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSwept() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsSwept()
```

C#

```
System.bool IsSwept()
```

C++/CLI

```
System.bool IsSwept();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if surface is a swept surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface.IsSwept](#).

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IModeler.CreateSweptSurface Method](#)
[IModeler.CreateSweptSurface Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsTorus Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : IsTorus Method (ISurface)

Gets whether the surface is a torus.

.NET Syntax

Visual Basic (Declaration)

```
Function IsTorus() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Boolean  
  
value = instance.IsTorus()
```

C#

```
System.bool IsTorus()
```

C++/CLI

```
System.bool IsTorus();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the surface is a torus, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::IsTorus](#).

Example

[Get Parameters of Toroidal Surface \(VBA\)](#)

See Also

[ISurface Interface](#)
[ISurface Members](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[ISurface::ITorusParams Property](#)
[ISurface::TorusParams Property](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ITorusParams Property (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : ITorusParams Property (ISurface)

Gets the parameters of a toroidal surface.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ITorusParams As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Double  
  
value = instance.ITorusParams
```

C#

```
System.double ITorusParams {get;}
```

C++/CLI

```
property System.double ITorusParams {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the parameters of a toroidal surface

Visual Basic for Applications (VBA) Syntax

See [Surface::ITorusParams](#).

Remarks

Returns an array of 8 double values:

center.x

center.y

center.z

axis.x

axis.y

axis.z

major radius - the distance between the center of torus and the center of revolved circle

minor radius - the radius of the revolved circle

NOTES:

- The real major radius (the outer radius) of the torus is major radius + minor radius.
- The center, major radius, and minor radius are in meters.
- Possible values that indicate a type of self-intersecting torus:
 - Apple - when the major radius is positive and less than or equal to the minor radius.
 - Lemon - when the major radius is negative and the sum of the radii is positive.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface.TorusParams Property](#)
[ISurface.IsTorus Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

MakIsoCurve Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : MakIsoCurve Method (ISurface)

Creates an untrimmed curve using the specified surface parameter.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function MakeIsoCurve( _  
    ByVal UorV As System.Boolean, _  
    ByVal UvValue As System.Double _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISurface  
Dim UorV As System.Boolean  
Dim UvValue As System.Double  
Dim value As System.Object  
  
value = instance.MakeIsoCurve(UorV, UvValue)
```

C#

```
System.object MakeIsoCurve(  
    System.bool UorV,  
    System.double UvValue  
)
```

C++/CLI

```
System.Object^ MakeIsoCurve(  
& System.bool UorV,  
& System.double UvValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UorV
True to specify v, false to specify u
UvValue
U or v value

Return Value

[Curve](#)

 Visual Basic for Applications (VBA) Syntax

See [Surface::MakeIsoCurve](#).

 See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IMakeIsoCurve Method](#)
[ISurface::IMakeIsoCurves Method](#)
[ISurface::MakeIsoCurves Method](#)
[ISurface::IGetMakeIsoCurvesCount Method](#)

 Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

MakeIsoCurve2 Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : MakeIsoCurve2 Method (ISurface)

Creates an untrimmed curve on a surface using the specified u or v surface function parameter.

 .NET Syntax

Visual Basic (Declaration)

```
Function MakeIsoCurve2( _  
    ByVal UorV As System.Boolean, _  
    ByRef UvValue As System.Double _  
    ) As Curve
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim UorV As System.Boolean  
Dim UvValue As System.Double  
Dim value As Curve  
  
value = instance.MakeIsoCurve2(UorV, UvValue)
```

C#

```
Curve MakeIsoCurve2(  
    System.bool UorV,  
    out System.double UvValue  
)
```

C++/CLI

```
Curve^ MakeIsoCurve2(  
& System.bool UorV,  
& [Out] System.double UvValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UorV
True to specify the surface function's v parameter in *UvValue*, false to specify its u parameter
UvValue

If <i>UorV</i> is...	Then <i>UvValue</i> is the surface function's...
True	V parameter
False	U parameter

Return Value

[Curve](#)

Visual Basic for Applications (VBA) Syntax

See [Surface::MakeIsoCurve2](#).

Example

[Create Trimmed Curve \(VBA\)](#)
[Create Trimmed Curve \(VB.NET\)](#)
[Create Trimmed Curve \(C#\)](#)

Remarks

This method supersedes the now obsolete `ISurface::MakeIsoCurve` by normalizing `UvValue` when it exceeds a specific value.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::MakeIsoCurves Method \(\)](#)
[ISurface::IMakeIsoCurves Method \(\)](#)
[ISurface::IGetMakeIsoCurvesCount Method \(\)](#)
[ISurface::IMakeIsoCurve Method \(\)](#)
[ICurve::CreateTrimmedCurve2 Method \(\)](#)

Availability

SOLIDWORKS 2013 SP05, Revision Number 21.5

SOLIDWORKS API Help

[Send comments](#) on this topic.

MakeIsoCurves Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : MakeIsoCurves Method (ISurface)

Creates curves on a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeIsoCurves( _  
    ByVal UvRange As System.Object, _  
    ByVal Dir As System.Object, _  
    ByVal Angle As System.Double, _  
    ByVal Tol As System.Double, _  
    ByRef Curves As System.Object, _  
    ByRef CurveBounds As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim UvRange As System.Object  
Dim Dir As System.Object  
Dim Angle As System.Double  
Dim Tol As System.Double  
Dim Curves As System.Object  
Dim CurveBounds As System.Object  
Dim value As System.Boolean
```

```
value = instance.MakeIsoCurves(UvRange, Dir, Angle, Tol, Curves, CurveBounds)
```

C#

```
System.bool MakeIsoCurves(  
    System.object UvRange,  
    System.object Dir,  
    System.double Angle,  
    System.double Tol,
```

```
    out System.object Curves,  
    out System.object CurveBounds  
)
```

C++/CLI

```
System.bool MakeIsoCurves(  
& System.Object^ UvRange,  
& System.Object^ Dir,  
& System.double Angle,  
& System.double Tol,  
& [Out] System.Object^ Curves,  
& [Out] System.Object^ CurveBounds  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UvRange

Array of 4 doubles indicating the range of the surface to use (see **Remarks**)

Dir

Array of 3 doubles indicating the direction of projection on the surface (see **Remarks**)

Angle

Angle relative to Dir where to create the curves

Tol

Tolerance of the curves

Curves

Array of [curves](#)

CurveBounds

Array of doubles of size 2*Curves indicating the parametric bounds of the curve

Return Value

True if the curves are created on the surface, false if not

Visual Basic for Applications (VBA) Syntax

See [Surface::MakeIsoCurves](#).

Remarks

The uvRange argument is an array of 4 doubles indicating the minimum and maximum U and V values:

[*u_min*, *u_max*, *v_min*, *v_max*]

The dir argument is an array of 3 doubles representing the unit vector:

[*x*, *y*, *z*]

See Also

[ISurface Interface](#)

[ISurface Members](#)

[ISurface::IMakeIsoCurves Method](#)

[ISurface::IMakeIsoCurve Method](#)

[ISurface::IGetMakeIsoCurvesCount Method](#)

[ISurface::MakeIsoCurve Method](#)

Availability

SOLIDWORKS 2004 SP2, Revision Number 12.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

Parameterization Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : Parameterization Method (ISurface)

Obsolete. Superseded by [ISurface::Parameterization2](#).

.NET Syntax

Visual Basic (Declaration)

Function Parameterization() As System.Object

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Object
```

```
value = instance.Parameterization()
```

C#

```
System.Object Parameterization()
```

C++/CLI

```
System.Object^ Parameterization();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of doubles describing the parameterization of the surface

Visual Basic for Applications (VBA) Syntax

See [Surface::Parameterization](#).

Remarks

The return value is the following array of 11 doubles:

$[uRange[2], vRange[2], uBoundType, vBoundType, uProps[2], vProps[2], numProps]$

where:

$uRange[2]$ = array of two doubles describing the low and high range, respectively, for U.

$vRange[2]$ = array of two doubles describing the low and high range, respectively, for V.

$uBoundType$ = two integers packed into a double. Each integer describes the behavior at the start and at the end of the U range. Valid values and their meanings are:

13733 - Infinite

13734 - Extendable

13735 - Not Extendable

13701 - Periodic

13736 - Periodic, but not continuously differentiable across the boundary

13741 - Degenerate

If the behavior at the start of the U range is 'Degenerate', then the V parameter is degenerate when $U = uRange[0]$. The derivative of the parameterization function with respect to v is 0 whenever $u = uRange[0]$, for all values of v, and the parameter curve corresponding to $u = uRange[0]$ degenerates to a single point. A parameterization can only degenerate at the end of its range, unless the surface is a type of Bsurface.

$vBoundType$ = two integers packed into a double. Each integer describes the behavior at the start and end of the V range. For valid values and their meanings, see description of $uBoundType$.

$uProps[2]$ = four integers packed into two doubles representing U parameterization properties. The $uProps$ array of two doubles will contain a total of one or more integer values with a maximum of four. Valid values and their meanings are:

13701 - Periodic parameterization.

13737 - All derivatives continuous.

13738 - All derivatives non necessarily continuous.

13739 - Linear. Corresponding parameter is proportional to the distance along a straight line. Straight line corresponds to a constant value of the other parameter.

13740 - Circular. Corresponding parameter represents an angle around a circle. Circle corresponds to a constant value of the other parameter.

13746 - Bounds at the ends of the parameter range are coincident.

$vProps[2]$ = four integers packed into two doubles representing V parameterization properties. The $vProps$ array of two doubles will contain a total of one or more integer values with a maximum of four. For valid values and their meanings, see description of $uProps$.

numProps = two integers packed into a double. Each integer represents the number of properties returned in *uProps* and *vProps*, respectively.

If a surface is periodic in one direction (that is, cylinder, cone, torus (apple and lemon shapes), spheres), then U is the periodic direction.

The -10000 to 10000 parameter range is the modeling limit for the modeler. This means that the values are effectively infinite in size. To get the parameter range from a given face, use [IFace2::GetBox](#) or [IFace2::!GetBox](#), [ISurface::GetClosestPointOn](#) or [ISurface::!GetClosestPointOn](#), and [ISurface::ReverseEvaluate](#) or [ISurface::!ReverseEvaluate](#). These methods provide a true parameter range that the face is using on the surface.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::!Parameterization Method](#)
[ISurface::Evaluate Method](#)
[ISurface::EvaluateAtPoint Method](#)
[ISurface::!Evaluate Method](#)
[ISurface::!EvaluateAtPoint Method](#)

SOLIDWORKS API Help

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Parameterization2 Method (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : Parameterization2 Method (ISurface)

Gets the parameterization of the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function Parameterization2() As SurfaceParameterizationData
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As SurfaceParameterizationData  
  
value = instance.Parameterization2()
```

C#

```
SurfaceParameterizationData Parameterization2()
```

C++/CLI

```
SurfaceParameterizationData^ Parameterization2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

An [ISurfaceParameterizationData](#) object

Visual Basic for Applications (VBA) Syntax

See [Surface::Parameterization2](#).

Example

[Get B-Spline Surface Parameterization Data \(C#\)](#)
[Get B-Spline Surface Parameterization Data \(VB.NET\)](#)
[Get B-Spline Surface Parameterization Data \(VBA\)](#)

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::!Parameterization Method](#)
[ISurface::Evaluate Method](#)
[ISurface::EvaluateAtPoint Method](#)
[!Evaluate Method](#)
[ISurface::!EvaluateAtPoint Method](#)
[ISurface::GetBSurfParams3 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

PlaneParams Property (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : PlaneParams Property (ISurface)

Gets the parameters of a planar surface.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property PlaneParams As System.Object

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Object  
  
value = instance.PlaneParams
```

C#

```
System.Object PlaneParams {get;}
```

C++/CLI

```
property System.Object^ PlaneParams {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the parameters of a planar surface

Visual Basic for Applications (VBA) Syntax

See [Surface::PlaneParams](#).

Example

[Create Infinite Plane \(VBA\)](#)
[Get Angle of Hole Not Normal to a Face \(VBA\)](#)
[Get Normal of Planar Surface \(VBA\)](#)
[Get Parameters of Planar Surface \(VBA\)](#)

Remarks

Returns an array of 6 double values:

normal.x
normal.y
normal.z
rootPoint.x
rootPoint.y
rootPoint.z

The *rootPoint* values are in meters.

See Also

[ISurface Interface](#)
[ISurface Members](#)

ReverseEvaluate Method (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : ReverseEvaluate Method (ISurface)

Gets the UV parameters at the specified location, which must be on the surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ReverseEvaluate( _  
    ByVal PositionX As System.Double, _  
    ByVal PositionY As System.Double, _  
    ByVal PositionZ As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim PositionX As System.Double  
Dim PositionY As System.Double  
Dim PositionZ As System.Double  
Dim value As System.Object  
  
value = instance.ReverseEvaluate(PositionX, PositionY, PositionZ)
```

C#

```
System.Object ReverseEvaluate(  
    System.double PositionX,  
    System.double PositionY,  
    System.double PositionZ  
)
```

C++/CLI

```
System.Object^ ReverseEvaluate(  
& System.double PositionX,  
& System.double PositionY,  
& System.double PositionZ  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PositionX
X position on the surface

PositionY
Y position on the surface

PositionZ
Z position on the surface

Return Value

Array of doubles describing the UV parameters

Visual Basic for Applications (VBA) Syntax

See [Surface::ReverseEvaluate](#).

Remarks

At parametric singularities, such as sphere poles, the non-degenerate parameter is returned as the lowest value in its range. To determine the UV range, call [ISurface::Parameterization](#) or [ISurface::IParameterization](#).

[IFace2::ReverseEvaluate](#) and [IFace2::IReverseEvaluate](#) return corrected UV parameters for periodic surfaces; thus, you should use either of these methods when dealing with periodic surfaces.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::IReverseEvaluate Method](#)
[ISurface::Evaluate Method](#)
[ISurface::EvaluateAtPoint Method](#)
[ISurface::IEvaluate Method](#)
[ISurface::IEvaluateAtPoint Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SphereParams Property (ISurface)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : SphereParams Property (ISurface)

Gets the parameters of a spherical surface.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SphereParams As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Object  
  
value = instance.SphereParams
```

C#

```
System.Object SphereParams {get;}
```

C++/CLI

```
property System.Object^ SphereParams {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of doubles describing the parameters of a spherical surface

Visual Basic for Applications (VBA) Syntax

See [Surface::SphereParams](#).

Remarks

Returns an array of 4 doubles:

center.x

center.y

center.z

radius

The *center* and *radius* values are in meters.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::ISphere Method](#)
[ISurface::ISphereParams Property](#)

TorusParams Property (ISurface)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISurface Interface](#) : TorusParams Property (ISurface)

Gets the parameters of a toroidal surface.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property TorusParams As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISurface  
Dim value As System.Object  
  
value = instance.TorusParams
```

C#

```
System.object TorusParams {get;}
```

C++/CLI

```
property System.Object^ TorusParams {  
    System.Object^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Surface::TorusParams](#).

Example

[Get Parameters of Toroidal Surface \(VBA\)](#)

Remarks

Returns an array of 8 double values:

center.x

center.y

center.z

axis.x

axis.y

axis.z

major radius - the distance between the center of torus and the center of revolved circle

minor radius - the radius of the revolved circle

NOTES:

- The real major radius (the outer radius) of the torus is major radius + minor radius.
- The center, major radius, and minor radius are in meters.
- Possible values that indicate a type of self-intersecting torus:
 - Apple - when the major radius is positive and less than or equal to the minor radius.
 - Lemon - when the major radius is negative and the sum of the radii is positive.

See Also

[ISurface Interface](#)
[ISurface Members](#)
[ISurface::ITorusParams Property](#)
[ISurface::ITorus Method](#)